Stacking galaxies reveals a universal baryon fountain

Averaging many simulated galaxy ecosystems uncovers smooth, regular gas flows matching observations.

arXiv:2609.21374 — Stacked baryon cycles in cosmological hydrodynamic simulations

Authors: Chengyu Ma, Yangyao Chen, Enci Wang, Houjun Mo, Huiyuan Wang, Tao Wang, Qirong Yuan, Min Du et al.

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Background

Galaxies grow by cycling gas in and out, forming fountain-like flows that link them to their halos. Individual ecosystems are too diverse and dynamic to reveal the underlying pattern directly.

> **Baryon cycles were inferred piecemeal from scattered evidence; no clean, unified picture of the flow geometry existed.**

What this paper is selling

The authors stack gas velocity and property fields across galaxy ensembles, averaging away idiosyncrasies. This exposes strikingly regular fountain patterns and yields a compact parameterization of the cycle.

Key results to remember

- Stacking suppresses galaxy-to-galaxy scatter, revealing coherent inflow and bipolar outflow cones.
- A simple fountain model reproduces the stacked velocity field with small residuals.
- The same fountain geometry persists across stellar mass and cosmic time, scaling with halo size.

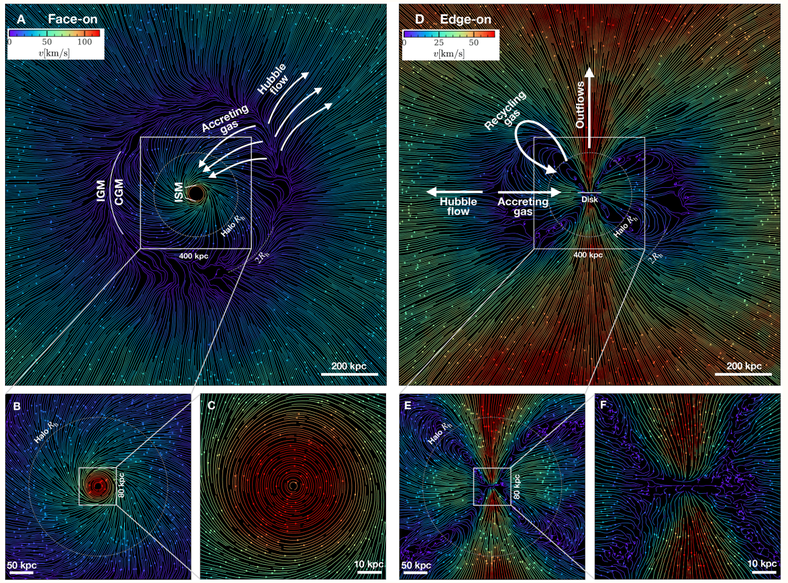


Figure 1 | Stacked streamlines show a smooth, coherent velocity field with clear inflow and outflow around typical star-forming galaxies.

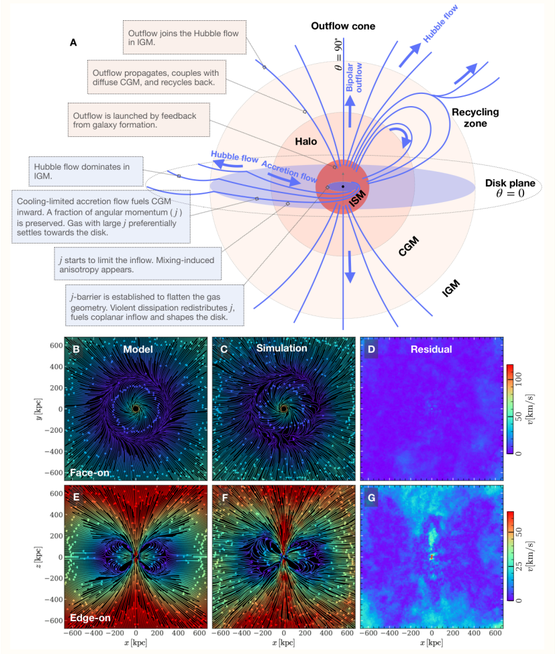


Figure 4 | A schematic fountain model matches the simulated stacked velocity field, compressing the baryon cycle into a few parameters.